

UNIT – 5
ASSESSMENT – 3
SOLUTION

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1.The system can be decomposed into the following components:

Patient Input



Data Preprocessing



Feature Extraction



AI Model



Disease Prediction

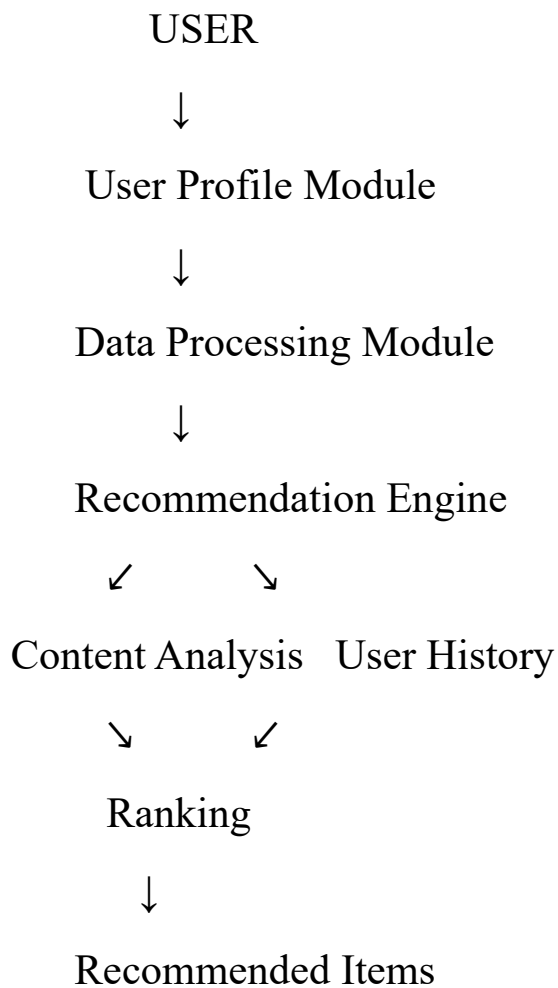


Result Display

Components

1. **Input Module** – collects patient symptoms.
2. **Preprocessing Module** – converts symptoms into usable features.
3. **Feature Extraction** – identifies important symptoms.
4. **AI Model** – analyzes the features.
5. **Prediction Module** – determines the possible disease.
6. **Output Module** – displays the prediction.

2. By observing inputs and outputs, we can reconstruct the following architecture:

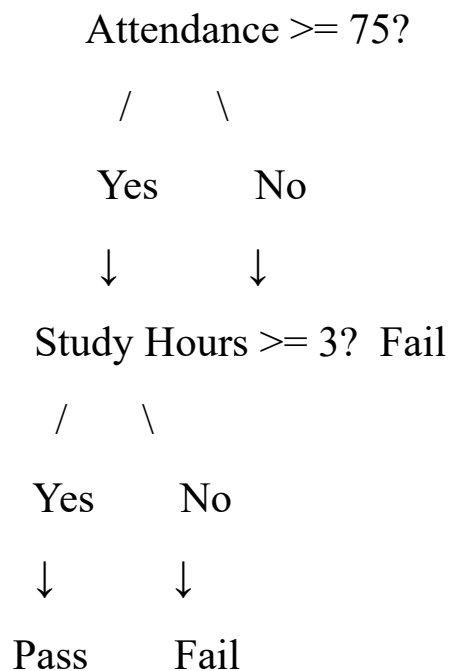


Reconstructed Modules

1. User Input
2. Profile Processing
3. Content Analysis
4. Recommendation Engine
5. Ranking
6. Output

3. Solution

From the observed behaviour, we can identify a simple **Decision Tree-like rule-based model**.



4. Solution

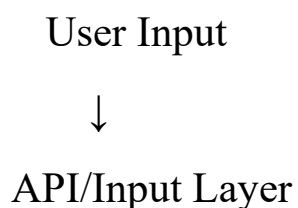
The system receives multiple transactions and analyzes them.

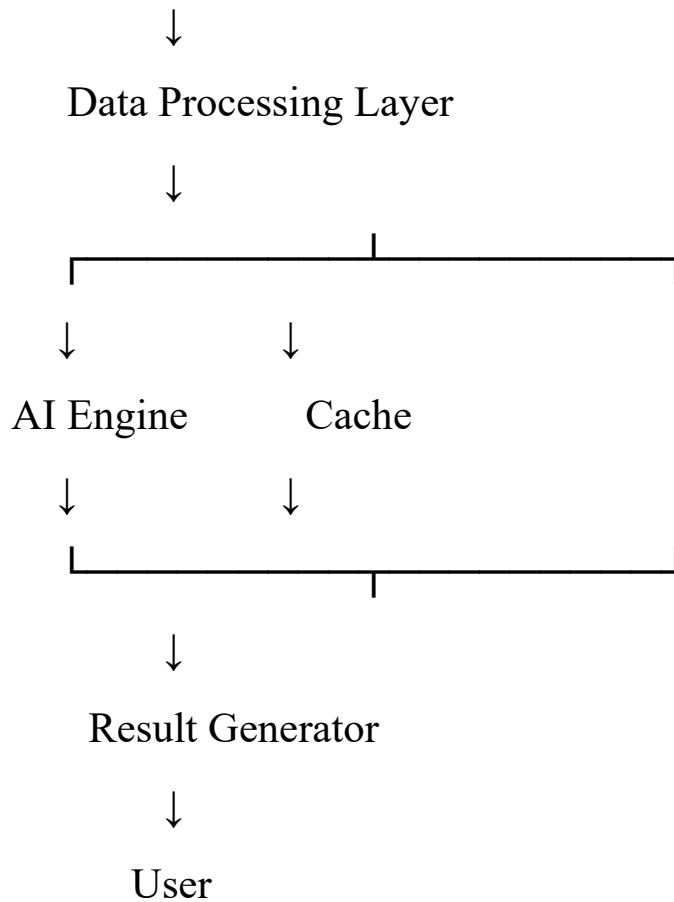
For demonstration, we will calculate a simple transaction risk score.

Risk conditions

- High transaction amount → higher risk
- Unknown location → higher risk
- Multiple suspicious conditions → fraud alert

5. The system can be redesigned as:





Improvements

Existing System	Proposed System
Single processing	Distributed processing
Sequential execution	Parallel execution
No caching	Result caching
Static model	Continuous learning
Limited scalability	Scalable architecture